CAVI for the Normal–Gamma Model

Variational updates and the ELBO

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Purpose. These notes follow the normal–Gamma example immediately after typed Slide 21 in the annotated Variational Inference lecture notes. We derive the updates for $q_\mu(\mu)$ and $q_\tau(\tau)$, calculate the expectations needed to implement them, and explain why the ELBO cannot decrease after an exact CAVI update. Throughout, $x = (x_1, \dots, x_n)$ is observed and held fixed. The Gamma distribution uses the shape–rate parameterisation.

1 The model and its densities

For $i = 1, \dots, n$, conditionally independently, the distributions and their densities are:

$$\begin{aligned} x_i \mid \mu, \tau &\sim \text{Normal}\left(\mu, \frac{1}{\tau}\right), \\ p(x_i \mid \mu, \tau) &= \sqrt{\frac{\tau}{2\pi}} \exp\left\{-\frac{\tau}{2}(x_i - \mu)^2\right\}, \quad x_i \in \mathbb{R}. \end{aligned} \tag{1}$$

$$\begin{aligned} \mu \mid \tau &\sim \text{Normal}\left(\mu_0, \frac{1}{\lambda_0 \tau}\right), \\ p(\mu \mid \tau) &= \sqrt{\frac{\lambda_0 \tau}{2\pi}} \exp\left\{-\frac{\lambda_0 \tau}{2}(\mu - \mu_0)^2\right\}, \quad \mu \in \mathbb{R}. \end{aligned} \tag{2}$$

$$\begin{aligned} \tau &\sim \text{Gamma}(a_0, b_0), \\ p(\tau) &= \frac{b_0^{a_0}}{\Gamma(a_0)} \tau^{a_0 - 1} e^{-b_0 \tau}, \quad \tau > 0. \end{aligned} \tag{3}$$

Here τ is the precision and $\lambda_0, a_0, b_0 > 0$. Multiplying the n conditional densities gives

$$p(x \mid \mu, \tau) = \left(\frac{\tau}{2\pi}\right)^{n/2} \exp\left\{-\frac{\tau}{2} \sum_{i=1}^{n} (x_i - \mu)^2\right\}. \tag{4}$$

Thus the joint density factorises as

$$\boxed{p(x, \mu, \tau) = p(x \mid \mu, \tau)\, p(\mu \mid \tau)\, p(\tau).} \tag{5}$$

2 Mean-field approximation and CAVI update rule

We approximate the posterior by

$$p(\mu, \tau \mid x) \approx q(\mu, \tau) = q_\mu(\mu) q_\tau(\tau). \tag{6}$$

The CAVI updates, each using the *current* other factor, are

$$\log q_\mu^*(\mu) = \mathbb{E}_{q_\tau}[\log p(x, \mu, \tau)] + C_\mu, \tag{7}$$

$$\log q_\tau^*(\tau) = \mathbb{E}_{q_\mu}[\log p(x, \mu, \tau)] + C_\tau. \tag{8}$$

C_μ denotes a term independent of μ and C_τ a term independent of τ; the normalising constants make the updated factors probability densities.

3 First update: derive $q_\mu^*(\mu)$ in full

Start from (7) and expand the joint density:

$$\log q_\mu^*(\mu) = \mathbb{E}_{q_\tau}[\log p(x \mid \mu, \tau) + \log p(\mu \mid \tau) + \log p(\tau)] + C_\mu. \tag{9}$$

Step 1: Write down the log-likelihood.

$$\log p(x \mid \mu, \tau) = \log \prod_{i=1}^{n} \left[\sqrt{\frac{\tau}{2\pi}} e^{-\frac{\tau}{2}(x_i - \mu)^2} \right] \tag{10}$$

$$= \sum_{i=1}^{n} \left[\frac{1}{2} \log \left(\frac{\tau}{2\pi} \right) - \frac{\tau}{2}(x_i - \mu)^2 \right] \tag{11}$$

$$= \frac{n}{2} \log \left(\frac{\tau}{2\pi} \right) - \frac{\tau}{2} \sum_{i=1}^{n} (x_i - \mu)^2. \tag{12}$$

Step 2: Write down the log-prior for $\mu \mid \tau$.

$$\log p(\mu \mid \tau) = \log \left[\sqrt{\frac{\lambda_0 \tau}{2\pi}} e^{-\frac{\lambda_0 \tau}{2}(\mu - \mu_0)^2} \right] \tag{13}$$

$$= \boxed{\frac{1}{2} \log \left(\frac{\lambda_0 \tau}{2\pi} \right) - \frac{\lambda_0 \tau}{2}(\mu - \mu_0)^2.} \tag{14}$$

Step 3: Write down the log-prior for τ.

$$\log p(\tau) = a_0 \log b_0 - \log \Gamma(a_0) + (a_0 - 1) \log \tau - b_0 \tau. \tag{15}$$

Step 4: Substitute all three pieces.

$$\log q_\mu^*(\mu) = \mathbb{E}_{q_\tau} \left[\frac{n}{2} \log \left(\frac{\tau}{2\pi} \right) - \frac{\tau}{2} \sum_{i=1}^{n} (x_i - \mu)^2 \right. \tag{16}$$

$$\left. + \frac{1}{2} \log \left(\frac{\lambda_0 \tau}{2\pi} \right) - \frac{\lambda_0 \tau}{2}(\mu - \mu_0)^2 + \log p(\tau) \right] + C_\mu. \tag{17}$$

Step 5: Keep only terms depending on μ. Terms involving only τ and fixed hyperparameters become constants after taking $\mathbb{E}_{q_\tau}$. Hence

$$\log q_\mu^*(\mu) = -\frac{\mathbb{E}_{q_\tau}[\tau]}{2} \left[\sum_{i=1}^{n} (x_i - \mu)^2 + \lambda_0 (\mu - \mu_0)^2 \right] + C_\mu. \tag{18}$$

Step 6: Expand the squares.

$$\sum_{i=1}^{n} (x_i - \mu)^2 = \sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2, \tag{19}$$

$$\lambda_0 (\mu - \mu_0)^2 = \lambda_0 \mu^2 - 2\lambda_0 \mu_0 \mu + \lambda_0 \mu_0^2. \tag{20}$$

Discarding terms independent of μ yields

$$\log q_\mu^*(\mu) = -\frac{\mathbb{E}_{q_\tau}[\tau]}{2} \left[(n + \lambda_0)\mu^2 - 2 \left(\sum_{i=1}^{n} x_i + \lambda_0 \mu_0 \right) \mu \right] + C_\mu. \tag{21}$$

Step 7: Complete the square and recognise the normal density. Let

$$m^* = \frac{\lambda_0 \mu_0 + \sum_{i=1}^{n} x_i}{\lambda_0 + n} = \frac{\lambda_0 \mu_0 + n\bar{x}}{\lambda_0 + n}. \tag{22}$$

Because $(n+\lambda_0)\mu^2 - 2(n+\lambda_0)m^*\mu = (n+\lambda_0)(\mu - m^*)^2 + \text{constant}$, we obtain

$$\log q_\mu^*(\mu) = -\frac{(n+\lambda_0)\mathbb{E}_{q_\tau}[\tau]}{2}(\mu - m^*)^2 + C_\mu. \tag{23}$$

Compare with $\log \text{Normal}(\mu; m, s^2) = -(\mu - m)^2/(2s^2) + \text{constant}$. Therefore

$$\boxed{q_\mu^*(\mu) = \text{Normal}(\mu; m^*, s^{2*}), \quad m^* = \frac{\lambda_0\mu_0 + n\bar{x}}{\lambda_0 + n}, \quad s^{2*} = \frac{1}{(\lambda_0+n)\mathbb{E}_{q_\tau}[\tau]}.} \tag{24}$$

Here $\text{Normal}(\mu; m, s^2)$ denotes the density evaluated at μ, with mean m and variance s^2.

Note (why completing the square identifies the normal distribution). The logarithm of a normal density has the form

$$\log f(\mu) = -\frac{(\mu - m)^2}{2s^2} + \text{constant}.$$

To connect this with Step 6, let $A = n + \lambda_0$ and $B = \sum_{i=1}^n x_i + \lambda_0\mu_0$. Then

$$\begin{aligned} A\mu^2 - 2B\mu &= A\left[\mu^2 - 2\frac{B}{A}\mu\right] \\ &= A\left[\left(\mu - \frac{B}{A}\right)^2 - \left(\frac{B}{A}\right)^2\right]. \end{aligned}$$

The last term does not depend on μ and can be absorbed into the constant. Thus the mean is $m^* = B/A$ and the inverse variance is $1/s^{2*} = A\mathbb{E}_{q_\tau}[\tau]$. *Completing the square is an algebraic step, not a separate CAVI update.*

4 Second update: derive $q_\tau^*(\tau)$ in full

From (8),

$$\log q_\tau^*(\tau) = \mathbb{E}_{q_\mu}[\log p(x \mid \mu, \tau) + \log p(\mu \mid \tau) + \log p(\tau)] + C_\tau \tag{25}$$

$$= \mathbb{E}_{q_\mu}\left[\frac{n}{2}\log\left(\frac{\tau}{2\pi}\right) - \frac{\tau}{2}\sum_{i=1}^n (x_i - \mu)^2 + \frac{1}{2}\log\left(\frac{\lambda_0\tau}{2\pi}\right)\right. \tag{26}$$

$$\left. - \frac{\lambda_0\tau}{2}(\mu - \mu_0)^2 + (a_0 - 1)\log\tau - b_0\tau\right] + C_\tau. \tag{27}$$

Now $\log\tau$ and τ depend on the variable being updated, so we must retain them. Taking expectations under q_μ,

$$\log q_\tau^*(\tau) = \left(\frac{n}{2} + \frac{1}{2} + a_0 - 1\right)\log\tau \tag{28}$$

$$- \tau\left[b_0 + \frac{1}{2}\sum_{i=1}^n \mathbb{E}_{q_\mu}[(x_i - \mu)^2] + \frac{\lambda_0}{2}\mathbb{E}_{q_\mu}[(\mu - \mu_0)^2]\right] + C_\tau. \tag{29}$$

A Gamma shape–rate density satisfies

$$g(\tau; a, b) = \frac{b^a}{\Gamma(a)}\tau^{a-1}e^{-b\tau}, \tag{30}$$

$$\log g(\tau; a, b) = (a - 1)\log\tau - b\tau + \text{constant}. \tag{31}$$

Comparing terms gives

$$a^* = a_0 + \frac{n+1}{2}, \tag{32}$$

$$b^* = b_0 + \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}_{q_\mu}[(x_i - \mu)^2] + \frac{\lambda_0}{2}\mathbb{E}_{q_\mu}[(\mu - \mu_0)^2]. \tag{33}$$

Thus

$$\boxed{q_\tau^*(\tau) = \text{Gamma}(\tau; a^*, b^*) \quad \text{(shape–rate)}.} \tag{34}$$

5 Calculate the expectations and iterate

We have derived the *form* of both updated densities. We now need to evaluate the expectations appearing in their parameters to make the updates numerically usable.

5.1 Update the normal factor q_μ

From Section 3,

$$\boxed{q_\mu^*(\mu) = \text{Normal}(\mu; m^*, s^{2*})} \tag{35}$$

with

$$\boxed{m^* = \frac{\lambda_0\mu_0 + \sum_{i=1}^n x_i}{\lambda_0 + n}}, \tag{36}$$

$$\boxed{s^{2*} = \frac{1}{(\lambda_0 + n)\mathbb{E}_{q_\tau}[\tau]}}. \tag{37}$$

What expectation do we need to calculate? The mean m^* is already explicit. To calculate s^{2*}, we need $\mathbb{E}_{q_\tau}[\tau]$. If the *current* Gamma factor is $q_\tau(\tau) = \text{Gamma}(\tau; a, b)$ (shape–rate), then

$$\boxed{\mathbb{E}_{q_\tau}[\tau] = \frac{a}{b}.} \tag{38}$$

Consequently the new normal variance is

$$\boxed{s^{2*} = \frac{b}{(\lambda_0 + n)a}.} \tag{39}$$

Here a, b are the *current* Gamma parameters, before updating the normal factor.

5.2 Update the Gamma factor q_τ

From Section 4,

$$\boxed{q_\tau^*(\tau) = \text{Gamma}(\tau; a^*, b^*)} \tag{40}$$

with

$$\boxed{a^* = a_0 + \frac{n+1}{2}}, \tag{41}$$

$$\boxed{\begin{aligned} b^* = b_0 + \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}_{q_\mu}[(x_i - \mu)^2] \\ + \frac{\lambda_0}{2}\mathbb{E}_{q_\mu}[(\mu - \mu_0)^2]. \end{aligned}} \tag{42}$$

What expectations do we need to calculate? The shape a^* is already explicit. The rate b^* requires the two squared-error expectations in (42). The first arises from the normal likelihood; the second arises from the conditional normal prior.

If the *current* normal factor is $q_\mu(\mu) = \text{Normal}(\mu; m, s^2)$, then for any constant c,

$$\mathbb{E}[(X - c)^2] = \text{Var}(X) + (\mathbb{E}[X] - c)^2. \tag{43}$$

Hence

$$\boxed{\mathbb{E}_{q_\mu}[(x_i - \mu)^2] = (x_i - m)^2 + s^2}, \tag{44}$$

$$\boxed{\mathbb{E}_{q_\mu}[(\mu - \mu_0)^2] = (m - \mu_0)^2 + s^2}. \tag{45}$$

For example, the first equality follows by expanding the square:

$$\begin{aligned}\mathbb{E}_{q_\mu}[(x_i - \mu)^2] &= x_i^2 - 2x_i\mathbb{E}_{q_\mu}[\mu] + \mathbb{E}_{q_\mu}[\mu^2] \\ &= x_i^2 - 2x_i m + (m^2 + s^2) \\ &= (x_i - m)^2 + s^2,\end{aligned}$$

because $\mathbb{E}_{q_\mu}[\mu] = m$ and $\mathbb{E}_{q_\mu}[\mu^2] = m^2 + s^2$. The same argument with x_i replaced by μ_0 gives the second equality.

Substituting these expectations into the Gamma rate formula yields

$$\boxed{\begin{aligned}b^* = b_0 &+ \frac{1}{2}\sum_{i=1}^{n}\left[(x_i - m)^2 + s^2\right] \\ &+ \frac{\lambda_0}{2}\left[(m - \mu_0)^2 + s^2\right].\end{aligned}} \tag{46}$$

All quantities on the right are available from the *current* normal factor.

5.3 One complete CAVI iteration

1. **Initialise:** choose valid Gamma parameters $a^{(0)}, b^{(0)} > 0$ (and initialise the normal factor if desired).

2. **Update q_μ:** at iteration t, calculate $\mathbb{E}_{q_\tau^{(t)}}[\tau] = a^{(t)}/b^{(t)}$; set

$$m^{(t+1)} = \frac{\lambda_0\mu_0 + \sum_i x_i}{n + \lambda_0}, \qquad s^{2(t+1)} = \frac{b^{(t)}}{(n + \lambda_0)a^{(t)}}.$$

3. **Update q_τ:** use this *new* normal distribution to calculate the two expectations in (44) and (45); then set

$$\begin{aligned}a^{(t+1)} &= a_0 + \frac{n+1}{2}, \\ b^{(t+1)} &= b_0 + \frac{1}{2}\sum_{i=1}^{n}\left[(x_i - m^{(t+1)})^2 + s^{2(t+1)}\right] \\ &\quad + \frac{\lambda_0}{2}\left[(m^{(t+1)} - \mu_0)^2 + s^{2(t+1)}\right].\end{aligned}$$

4. **Check convergence:** evaluate the ELBO with the updated factors. If its change is not yet sufficiently small, repeat the cycle.

Key distinction. The update *formulas* are derived once, but the variational parameters are recalculated at each iteration. In this example m^* and a^* are fixed by the data and prior; the normal variance and Gamma rate generally change across iterations. We are updating *distributions*, not sampling new values of μ and τ.

6 Calculating the ELBO for the normal–Gamma example

We now calculate the ELBO as in the lecture notes, by expanding the joint density and the two variational densities. In this section we use $\mu^* = m^*$ and $\sigma^{2*} = s^{2*}$ for the parameters of the updated normal factor. The expectations below are taken with respect to $q_\mu^*(\mu) q_\tau^*(\tau)$.

$$\begin{aligned}
\mathrm{ELBO}(q_\mu^*(\mu), q_\tau^*(\tau)) &= \mathrm{ELBO}(\mu^*, \sigma^{2*}, a^*, b^*) && (47)\\
&= \mathbb{E}_{\mu,\tau}\big[\log p(x_1, \ldots, x_n, \mu, \tau) - \log\big(q_\mu^*(\mu) q_\tau^*(\tau)\big)\big] && (48)\\
&= \mathbb{E}_{\mu,\tau}\left[\log\left\{\prod_{i=1}^{n} p(x_i \mid \mu, \tau)\, p(\mu \mid \tau)\, p(\tau)\right\}\right] \\
&\quad - \mathbb{E}_{\mu,\tau}[\log q_\mu^*(\mu)] - \mathbb{E}_{\mu,\tau}[\log q_\tau^*(\tau)]. && (49)
\end{aligned}$$

We compute these three expectations separately. The notation $\mathbb{E}_{\mu,\tau}$ means expectation under the *variational* product $q_\mu^* q_\tau^*$, not under the exact posterior.

6.1 Expected log normal variational density

Since $q_\mu^*(\mu) = \mathrm{Normal}(\mu; \mu^*, \sigma^{2*})$,

$$\begin{aligned}
\mathbb{E}_{\mu,\tau}[\log q_\mu^*(\mu)] &= \mathbb{E}_{\mu,\tau}\left[-\frac{1}{2}\log \sigma^{2*} - \frac{1}{2}\log(2\pi) - \frac{(\mu - \mu^*)^2}{2\sigma^{2*}}\right] && (50)\\
&= -\frac{1}{2}\log \sigma^{2*} - \frac{\mathbb{E}_\mu[(\mu - \mu^*)^2]}{2\sigma^{2*}} + \text{const} && (51)\\
&= -\frac{1}{2}\log \sigma^{2*} - \frac{\sigma^{2*}}{2\sigma^{2*}} + \text{const} && (52)\\
&= -\frac{1}{2}\log \sigma^{2*} + \text{const}. && (53)
\end{aligned}$$

Here $\mathbb{E}_\mu[(\mu - \mu^*)^2] = \sigma^{2*}$, because μ^* and σ^{2*} are the mean and variance of q_μ^*. In these lines, const collects terms that do not depend on μ^* or σ^{2*} (including $-\frac{1}{2}\log(2\pi) - \frac{1}{2}$). Usually it is enough to calculate the ELBO up to a constant.

6.2 Expected log Gamma variational density

For $q_\tau^*(\tau) = \mathrm{Gamma}(\tau; a^*, b^*)$, with b^* the *rate*,

$$\begin{aligned}
\mathbb{E}_{\mu,\tau}[\log q_\tau^*(\tau)] &= \mathbb{E}_{\mu,\tau}[a^* \log b^* - \log\Gamma(a^*) + (a^* - 1)\log\tau - b^*\tau] && (54)\\
&= a^* \log b^* - \log\Gamma(a^*) + (a^* - 1)\mathbb{E}_\tau[\log\tau] - b^*\mathbb{E}_\tau[\tau] && (55)\\
&= a^* \log b^* - \log\Gamma(a^*) + (a^* - 1)[-\log b^* + \Psi(a^*)] - b^*\frac{a^*}{b^*} && (56)\\
&= \log b^* - \log\Gamma(a^*) + (a^* - 1)\Psi(a^*) - a^*. && (57)
\end{aligned}$$

Here $\Psi(z) = \frac{d}{dz}\log\Gamma(z)$ is the *digamma function*. We used

$$\mathbb{E}_\tau[\log\tau] = \Psi(a^*) - \log b^*, \qquad \mathbb{E}_\tau[\tau] = \frac{a^*}{b^*}.$$

6.3 Expected log joint model density

The remaining term in (49) is

$$\begin{aligned}
&\mathbb{E}_{\mu,\tau}\left[\log\left\{\prod_{i=1}^{n} p(x_i \mid \mu,\tau)p(\mu \mid \tau)p(\tau)\right\}\right] \\
&= \mathbb{E}_{\mu,\tau}\left[\frac{n}{2}\log\tau - \frac{n}{2}\log(2\pi) - \frac{\tau}{2}\sum_{i=1}^{n}(x_i-\mu)^2\right. \\
&\qquad \left. + \frac{1}{2}\log\lambda_0 + \frac{1}{2}\log\tau - \frac{1}{2}\log(2\pi) - \frac{\lambda_0\tau}{2}(\mu-\mu_0)^2 + \log p(\tau)\right]. \qquad (58)
\end{aligned}$$

As in the lecture notes, first collect the terms from the likelihood and the conditional prior for μ. This gives

$$\begin{aligned}
&\mathbb{E}_{\mu,\tau}\left[\frac{n}{2}\log\tau - \frac{n}{2}\log(2\pi) - \frac{\tau}{2}\sum_{i=1}^{n}(x_i-\mu)^2 + \frac{1}{2}\log\lambda_0 + \frac{1}{2}\log\tau\right. \\
&\qquad \left. - \frac{1}{2}\log(2\pi) - \frac{\lambda_0\tau}{2}(\mu-\mu_0)^2\right] + \text{const} \\
&= \mathbb{E}_{\mu,\tau}\left[\frac{n+1}{2}\log\tau - \frac{\tau}{2}\left\{\sum_{i=1}^{n}(x_i-\mu)^2 + \lambda_0(\mu-\mu_0)^2\right\}\right] + \text{const} \qquad (59) \\
&= \frac{n+1}{2}\mathbb{E}_\tau[\log\tau] - \frac{1}{2}\mathbb{E}_\tau[\tau]\left\{\sum_{i=1}^{n}\mathbb{E}_\mu[(x_i-\mu)^2] + \lambda_0\mathbb{E}_\mu[(\mu-\mu_0)^2]\right\} + \text{const} \qquad (60) \\
&= \frac{n+1}{2}[-\log b^* + \Psi(a^*)] \\
&\quad - \frac{a^*}{2b^*}\left\{\sum_{i=1}^{n}(\sigma^{2*} + \mu^{*2} - 2x_i\mu^* + x_i^2) + \lambda_0(\sigma^{2*} + \mu^{*2} - 2\mu_0\mu^* + \mu_0^2)\right\} + \text{const}. \qquad (61)
\end{aligned}$$

The third equality uses the mean-field factorisation: under $q_\mu^* q_\tau^*$, μ and τ are independent, so an expectation of τ times a function of μ separates into the product of two expectations. The squared-error expectations are the ones calculated in Section 5.

Finally, calculate the Gamma prior contribution separately:

$$\begin{aligned}
\mathbb{E}_\tau[\log p(\tau)] &= \mathbb{E}_\tau[a_0\log b_0 - \log\Gamma(a_0) + (a_0-1)\log\tau - b_0\tau] \qquad (62) \\
&= (a_0-1)\mathbb{E}_\tau[\log\tau] - b_0\mathbb{E}_\tau[\tau] + \text{const} \qquad (63) \\
&= (a_0-1)[-\log b^* + \Psi(a^*)] - b_0\frac{a^*}{b^*} + \text{const}. \qquad (64)
\end{aligned}$$

Here the constant in the final display is $a_0\log b_0 - \log\Gamma(a_0)$.

Putting the pieces together. Substitute the expected log joint term (the sum of (61) and (64)) and *subtract* the two expected log variational densities (53) and (57), exactly as prescribed by (49). This gives the ELBO calculation. When monitoring numerical convergence, keep the same constants across all iterations (or include them all); they cancel from differences of ELBO values.